

MODULE 13

Natural Language Processing NLP

AI with Data Science - Professional Learning Series

ARSHITH LMS | Professional Edition

Natural Language Processing NLP

What is NLP?

NLP enables machines to understand and process human language.

Key tasks:

Text Classification: sentiment, spam, topic

Named Entity Recognition: persons, places, orgs

Machine Translation: English to French

Question Answering: reading comprehension

Text Generation: GPT, story generation

Summarization: extract key points

Text Preprocessing

```
import re
import nltk
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer

nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')

def preprocess(text):
    text = text.lower()
    text = re.sub(r'^a-z\s', '', text)
    tokens = word_tokenize(text)
    stop_words = set(stopwords.words('english'))
    tokens = [t for t in tokens if t not in stop_words]
    lem = WordNetLemmatizer()
    tokens = [lem.lemmatize(t) for t in tokens]
    return ' '.join(tokens)
```

Text Vectorization

```
from sklearn.feature_extraction.text import TfidfVectorizer, CountVectorizer

# Bag of Words
cv = CountVectorizer(max_features=5000)
X = cv.fit_transform(texts)

# TF-IDF (Term Frequency-Inverse Document Frequency)
tfidf = TfidfVectorizer(max_features=5000, ngram_range=(1,2))
X = tfidf.fit_transform(texts)
```

```
# Word Embeddings (dense vectors)
from gensim.models import Word2Vec
model = Word2Vec(sentences, vector_size=100, window=5, min_count=1)
vector = model.wv['python'] # 100-dim vector for 'python'
```

Sentiment Analysis

```
from transformers import pipeline

# Using HuggingFace Transformers (pretrained BERT)
sentiment = pipeline('sentiment-analysis')
result = sentiment('The course was absolutely amazing!')
print(result) # [{'label': 'POSITIVE', 'score': 0.999}]

# Traditional ML approach
from sklearn.linear_model import LogisticRegression
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.pipeline import Pipeline

pipe = Pipeline([
    ('tfidf', TfidfVectorizer()),
    ('clf', LogisticRegression())
])
pipe.fit(X_train_text, y_train)
predictions = pipe.predict(X_test_text)
```

Transformers and BERT

Transformers revolutionised NLP in 2017.

Self-attention: each word attends to all others.

BERT (2018): Bidirectional Encoder Representations

GPT (2018): Generative Pre-trained Transformer

GPT-4, Claude, Gemini: latest LLMs

```
from transformers import BertTokenizer, BertForSequenceClassification
import torch
```

```
tokenizer = BertTokenizer.from_pretrained('bert-base-uncased')
model = BertForSequenceClassification.from_pretrained('bert-base-uncased',
num_labels=2)
```

```
inputs = tokenizer('Hello world!', return_tensors='pt')
outputs = model(**inputs)
```